ClimateScope: Global Weather Analysis

Project Report

by

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Project:

ClimateScope – Visualizing Global Weather Trends and Extreme Events

Date:

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1. Introduction

Climate change and weather variability have become increasingly important areas of study due to their impact on ecosystems, agriculture, and human life. Understanding climate patterns through data analysis helps in identifying trends, predicting future conditions, and supporting decision-making processes.

The ClimateScope project aims to analyze global weather data and present it through an interactive dashboard. The project focuses on identifying trends, seasonal variations, regional differences, and extreme weather events using data-driven techniques.

By combining data preprocessing, analysis, and visualization, this project transforms raw weather data into meaningful and actionable insights.

2. Dataset Description

- Dataset Name: Global Weather Repository
- Source: Kaggle
- Total Records: 125,000+
- Number of Features: 40+
- Coverage: Global
- Update Frequency: Daily

Key Variables Used

- Temperature (°C)
- Humidity (%)
- Precipitation (mm)
- Wind Speed (km/h)
- Atmospheric Pressure (mb)
- Timestamp (Date & Time)
- Geographic Data (Country, Latitude, Longitude)

The dataset provides comprehensive information about global weather conditions, making it suitable for climate analysis.

3. Methodology

The project follows a structured workflow consisting of data preprocessing, feature engineering, and analysis.

3.1 Data Preprocessing

- Removed duplicate records
- Handled missing values:
  - Numerical → replaced with mean
  - Categorical → filled with "Unknown"
- Converted timestamp into datetime format
- Extracted **year** and **month** from date
- Standardized units for consistency

3.2 Feature Engineering

A new feature called **season** was created to enable seasonal analysis.
Months were classified as:

- Winter: December – February
- Spring: March – May
- Summer: June – August
- Autumn: September – November

This helps in analyzing climate behavior across different seasons.

3.3 Data Aggregation

- Data was grouped by **country, year, and month**
- Monthly averages were calculated
- Aggregation helps reduce noise and improves clarity in analysis

3.4 Exploratory Data Analysis (EDA)

- Distribution analysis of temperature and precipitation
- Trend analysis over time
- Correlation between climate variables

- Identification of extreme weather events

4. Analysis and Key Insights

4.1 Climate Trends

- Temperature shows gradual variation over time
- Seasonal patterns are clearly visible
- Slight increasing trend observed in long-term temperature data

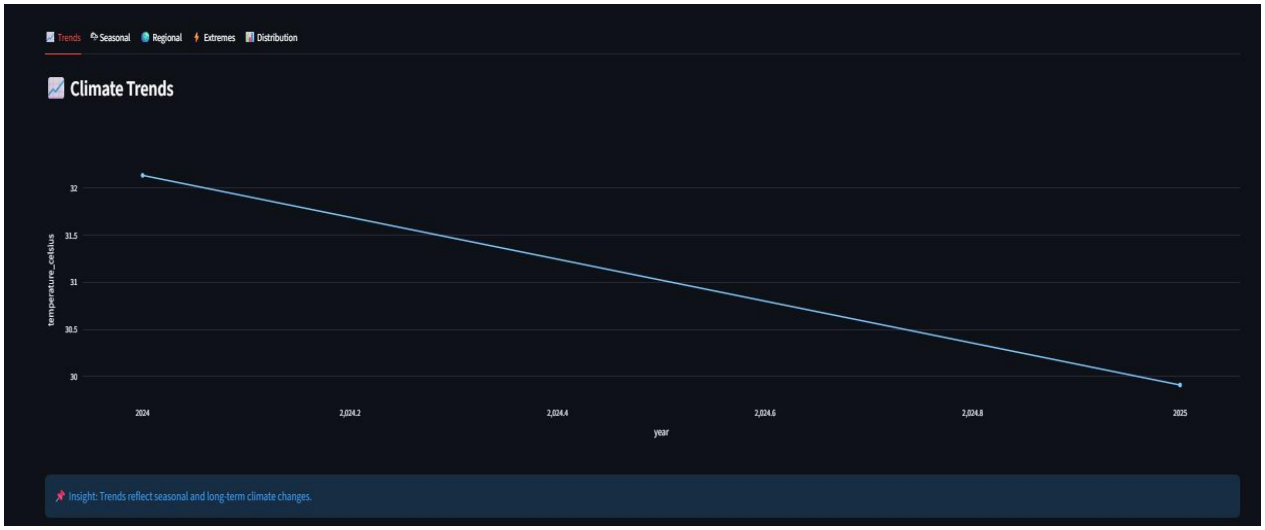

Figure 1: ClimateScope Dashboard Overview

The figure shows the main interface of the ClimateScope dashboard. It includes interactive filters such as country selection, year range, and season, along with key climate metrics like average temperature, humidity, and precipitation. The layout is organized into multiple sections for trends, seasonal analysis, regional comparison, extreme events, and distribution analysis, enabling users to explore global climate data efficiently.

4.2 Seasonal Patterns

- Summer consistently records the highest temperatures
- Winter shows the lowest temperatures
- Precipitation varies significantly across seasons
- Some regions show irregular rainfall patterns

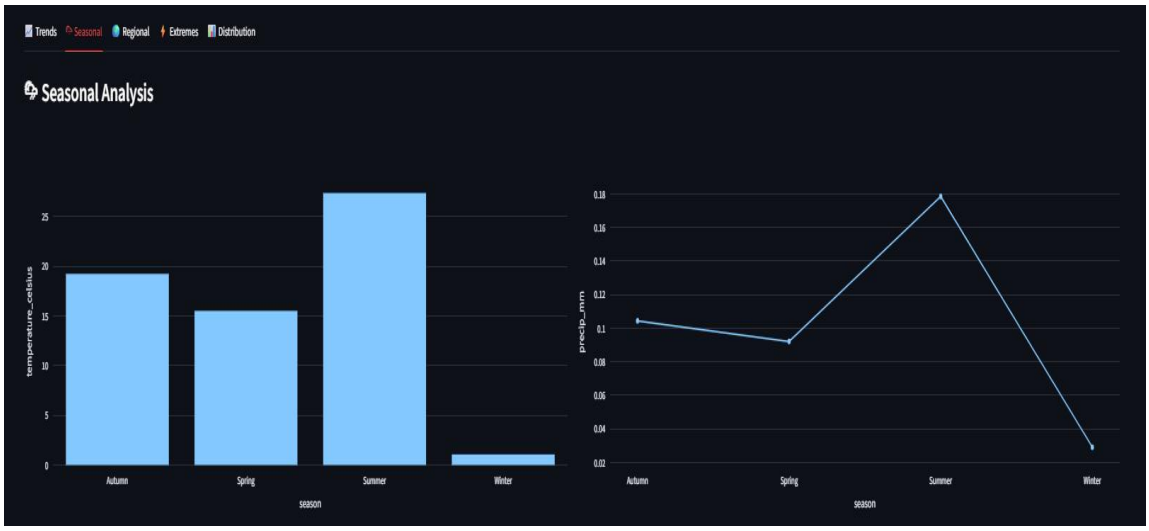

Figure 2: Seasonal Temperature and Precipitation Patterns

The figure presents seasonal variations in temperature and precipitation. It shows how different seasons influence climate conditions, with higher temperatures observed in summer and lower values in winter. Precipitation patterns vary across seasons, indicating regional climate differences.

4.3 Regional Comparison

- Climate conditions vary significantly across countries
- Coastal regions tend to have higher wind speeds
- Temperature distribution differs across geographic locations

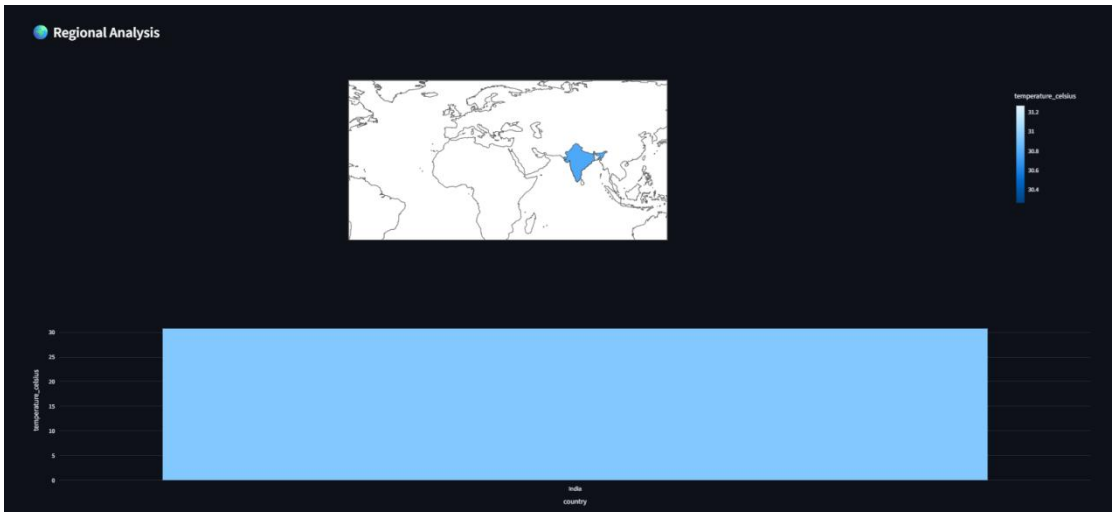

Figure 3: Regional Climate Distribution Map

This map visualizes the geographic distribution of climate variables across different countries. Color intensity represents variations in the selected metric, allowing users to compare regional climate conditions. The map provides insights into how weather patterns differ globally.

4.4 Extreme Weather Events

- Extreme events were identified using the 95th percentile threshold
- This ensures only the most significant values are considered

Important improvement:

Extreme values were calculated using filtered (region-specific) data instead of global data to ensure accurate and meaningful results.

- Extreme events are rare but important indicators of climate behavior
- They are concentrated in specific regions

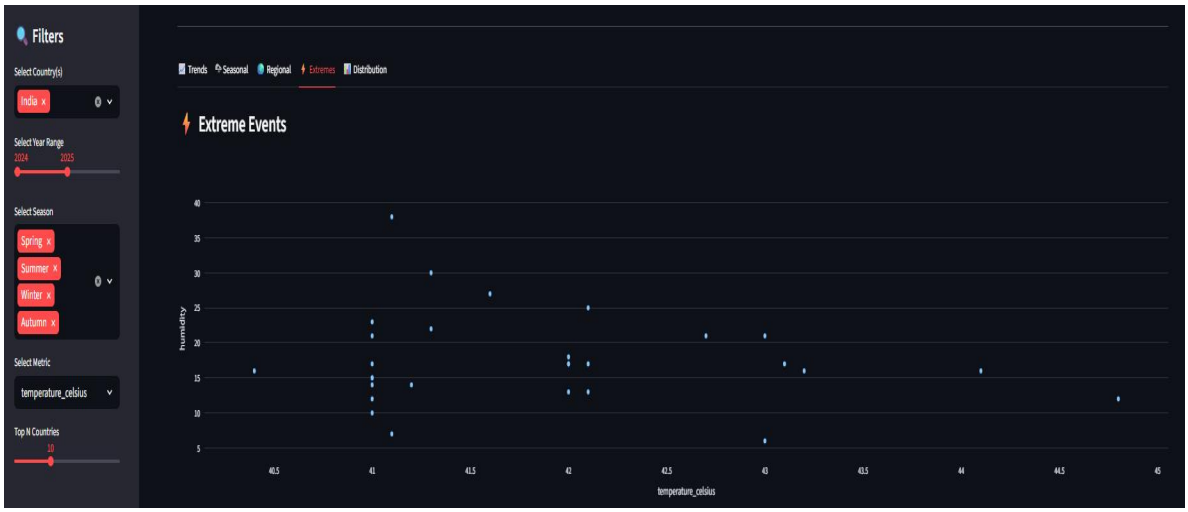

Figure 4: Regional Climate Distribution Map

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4.5 Correlation Analysis

- Weak to moderate relationship between temperature and humidity
- Pressure and wind speed show inverse relationship
- Climate variables do not follow simple linear relationships

This highlights the complexity of weather systems.

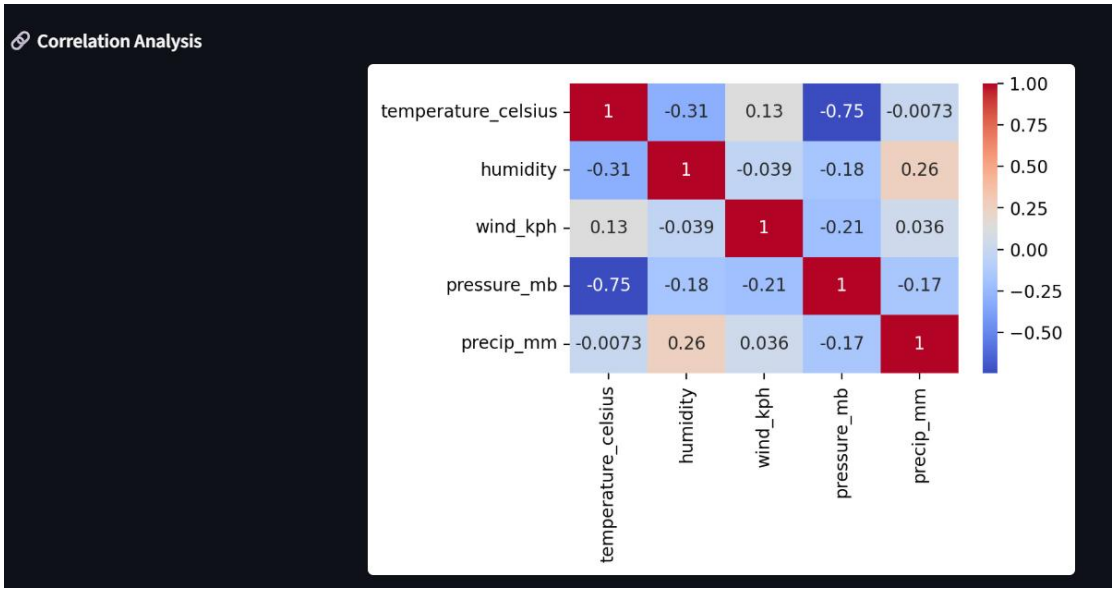

Figure 5: Extreme Weather Events Detection

This figure highlights extreme weather events identified using the 95th percentile threshold. The scatter plot shows outliers representing unusually high temperature or precipitation values. These extreme events are rare but significant for understanding climate risks.

5. Dashboard Implementation

An interactive dashboard was developed using:

- **Streamlit** for web interface
- **Plotly** for visualization

Dashboard Features

- Dynamic filters:
  - Country selection
  - Year range
  - Season selection
  - Metric selection
- Visualization sections:

- Climate Trends
- Seasonal Analysis
- Regional Comparison
- Extreme Weather Events
- Data Distribution
- Correlation Analysis

User Interaction

The dashboard allows users to:

- Explore data for specific countries
- Analyze seasonal patterns
- Compare regions
- Identify extreme weather conditions
- Understand relationships between variables

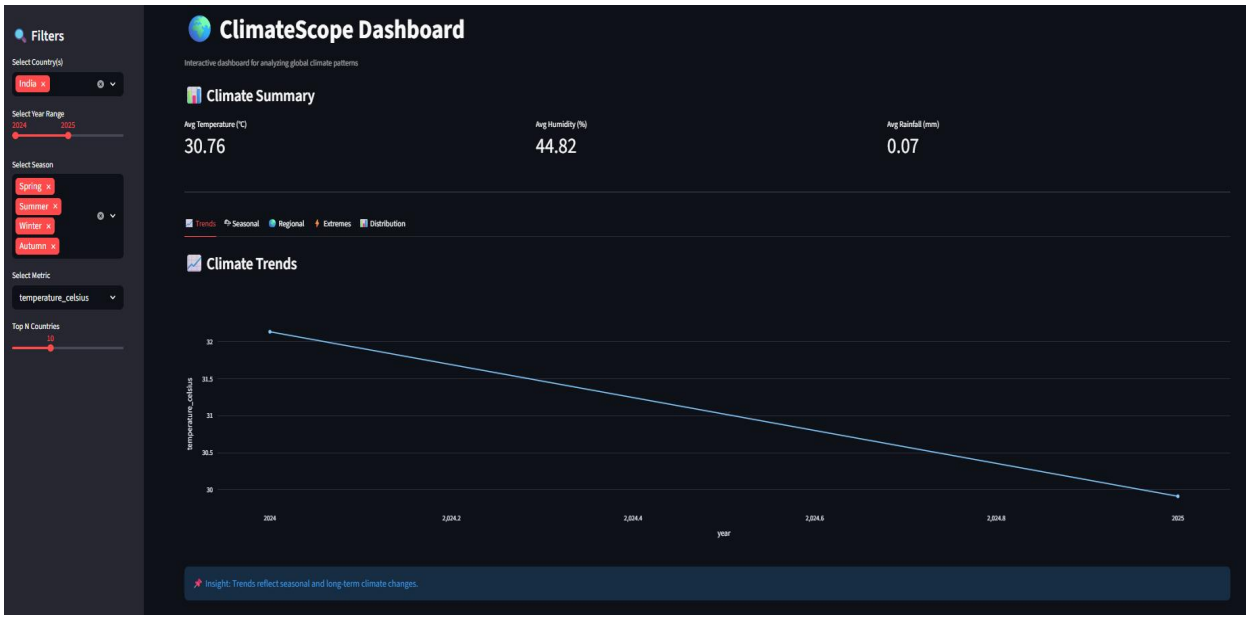

Figure 6 : ClimateScope Dashboard Overview

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6. Testing and Validation

The dashboard was tested to ensure accuracy, functionality, and stability.

Test Cases Performed

- Country filter functionality
- Year range selection
- Seasonal filtering
- Metric selection
- Map accuracy
- Extreme event detection
- Correlation visualization
- Empty selection handling

Results

- All filters function correctly
- Charts update dynamically
- Map shows accurate regions
- No crashes or runtime errors observed
- Dashboard performs efficiently

All test cases passed successfully, confirming that the dashboard is stable and reliable.

7. Challenges Faced

- Handling missing and inconsistent data

- Fixing incorrect regional mapping in visualizations
- Defining appropriate thresholds for extreme events
- Ensuring all visualizations respond correctly to filters

8. Improvements Made

- Replaced global data usage with filtered data
- Improved map accuracy
- Enhanced interactivity and user experience
- Optimized visualization clarity

9. Conclusion

The ClimateScope project successfully demonstrates how global weather data can be analyzed and visualized to generate meaningful insights.
The dashboard provides a user-friendly platform to explore climate patterns, seasonal trends, and extreme events. The project highlights the importance of data-driven analysis in understanding environmental changes.

10. Deployment

The ClimateScope dashboard was deployed using Streamlit Cloud to enable real-time access and usability without requiring local setup.The deployment process involved integrating the GitHub repository with Streamlit Cloud and configuring the application environment using the requirements file.

Live Application Link:
ClimateScope Dashboard • Streamlit

The deployed dashboard allows users to interactively explore climate data, apply filters, and analyze trends from any device with internet access.This deployment ensures scalability, accessibility, and ease of demonstration for stakeholders.

10. Future Enhancements

- Integration of machine learning models for prediction
- Real-time data updates
- Advanced anomaly detection
- Improved UI/UX design